

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ -M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{T}_z^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} M_{xxx} & 0 & 0 \\ -M_{xxx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & -M_{xxx} & 0 \end{bmatrix}$$

$$M_{xxx} = \mathbb{M}_{3a}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{yzx} & 0 & 0 \\ 0 & -M_{yzx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzx} = 3\mathbb{T}_u^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzx} & 0 & 0 \\ 0 & M_{yzx} & 0 \\ 0 & 0 & M_{xyz} \end{bmatrix}$$

$$M_{yzx} = \mathbb{T}_0^{(1)} - \mathbb{T}_u^{(2)}$$

$$M_{xyz} = \mathbb{T}_0^{(1)} + 2\mathbb{T}_u^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxxx} & 0 \\ 0 & 0 & 0 & 0 & -M_{xxxx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -M_{xxxx} \\ M_{xxxx} & -M_{xxxx} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -M_{xxxx} & 0 & 0 \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_{4b}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxxx} & M_{xxxxy} \\ 0 & 0 & 0 & 0 & -M_{xxxx} & -M_{xxxxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{yyzzx} & M_{xxxzx} \\ -M_{xxxzx} & M_{xxxzx} & 0 & -M_{yyzzx} & 0 & 0 \\ -M_{xxxxy} & M_{xxxxy} & 0 & -M_{xxxzx} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= -2\mathbb{T}_{3b}^{(1)} \\
M_{xxxy} &= 2\mathbb{T}_z^{(1)} - 2\mathbb{T}_z^{\alpha(1)} \\
M_{yyzz} &= -\mathbb{T}_z^{(1)} - 4\mathbb{T}_z^{\alpha(1)}
\end{aligned}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & M_{yyzz} & 0 \\ -M_{yyzz} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yyzz} = -\mathbb{T}_z^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & M_{xxzx} & M_{xxxy} \\ 0 & -M_{xxzx} & M_{xxxy} \\ 0 & 0 & M_{zzxy} \\ 0 & M_{yyzx} & 0 \\ M_{yyzz} & 0 & 0 \\ M_{xxzx} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= \mathbb{T}_{3b}^{(2)} \\
M_{xxxy} &= \mathbb{T}_z^{(3)} - \mathbb{T}_z^{\alpha(2)} \\
M_{zzxy} &= \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)} \\
M_{yyzz} &= \mathbb{T}_z^{(4)} - \mathbb{T}_z^{\alpha(2)}
\end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yyzx} & M_{yyzy} \\ M_{yyzy} & -M_{yyzy} & 0 & M_{yyzx} & 0 & 0 \\ M_{xyyx} & M_{xyyx} & M_{xyzz} & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{yyzz} &= \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} \\
M_{yyzy} &= \mathbb{T}_{3b}^{(3)} \\
M_{xyyx} &= \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} \\
M_{xyzz} &= \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}
\end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{xxzx} & M_{xxxy} & 0 & M_{xxxz} & M_{xxyx} \\ 0 & 0 & 0 & 0 & -M_{xxzx} & -M_{xxxy} & 0 & -M_{xxxz} & -M_{xxyx} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & -M_{zzxy} \\ 0 & 0 & 0 & 0 & M_{yyzx} & M_{yyzy} & 0 & M_{yyxz} & M_{yyzy} \\ -M_{yyzy} & M_{yyzy} & 0 & -M_{yyzx} & 0 & 0 & -M_{yyzx} & 0 & 0 \\ -\frac{M_{xxxy}}{2} - \frac{M_{xxyx}}{2} & \frac{M_{xxxy}}{2} + \frac{M_{xxyx}}{2} & 0 & -M_{xxzx} & 0 & 0 & -M_{xxzx} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= \mathbb{M}_{4b}^{(1)} - 2\mathbb{T}_{3b}^{(1)} + \mathbb{T}_{3b}^{(2)} \\
M_{xxxy} &= 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} \\
M_{xxxz} &= \mathbb{M}_{4b}^{(1)} - 2\mathbb{T}_{3b}^{(1)} - \mathbb{T}_{3b}^{(2)} \\
M_{xxyx} &= 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} \\
M_{zzxy} &= \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)} \\
M_{yyzz} &= -\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} \\
M_{yyzy} &= -\mathbb{M}_{4b}^{(1)} - 2\mathbb{T}_{3b}^{(1)} \\
M_{yyxz} &= -\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & M_{yzxz} & M_{yzxy} & 0 & M_{yzxz} & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & M_{yzxz} & 0 & 0 & M_{yzxz} & 0 & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzz} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzxz} = -\mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{yzxy} = \mathbb{T}_{3b}^{(3)}$$

$$M_{yzxz} = \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{xyxx} = \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)}$$

$$M_{xyzz} = \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & M_{xxxx} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ -M_{xxxx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{xxxx} & 0 & 0 \\ 0 & 0 & -M_{xxxx} \\ 0 & -M_{xxxx} & 0 \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_{4b}^{(1)}$$